



US 20250281812A1

(19) **United States**

(12) **Patent Application Publication**  
**Peterson**

(10) **Pub. No.: US 2025/0281812 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **EXOSKELETON ROBOTIC SYSTEM FOR  
GOLF-SWING TRAINING AND MEDICAL  
REHABILITATION**

(52) **U.S. Cl.**  
CPC ..... **A63B 69/3608** (2013.01)

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(21) Appl. No.: **18/601,738**

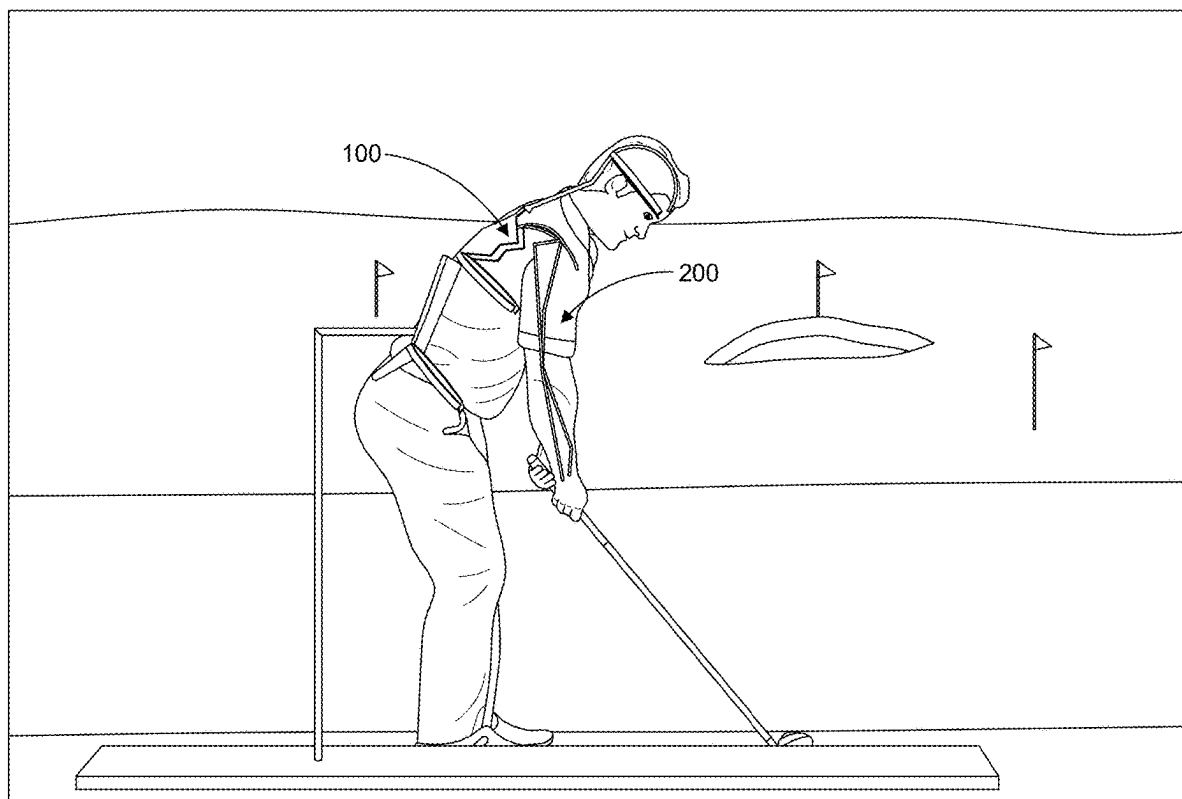
(22) Filed: **Mar. 11, 2024**

**Publication Classification**

(51) **Int. Cl.**  
**A63B 69/36** (2006.01)

(57) **ABSTRACT**

A robotic training device, system and method duplicates and guides a golfer through a perfect golf swing in several modes. The exoskeleton-like robotic device, and system utilizes a training method that employs a combination of either pneumatic, electric or Bowden cables with small motors and actuators, which embraces the golfer so he/she can experience the “feel” and body movements that make a perfect golf swing and learn to perform the motion without the training device. Similarly, the robot training device, system may assist patients in rehabilitation from injuries, stroke, chronic illness or other conditions involving muscular atrophy.



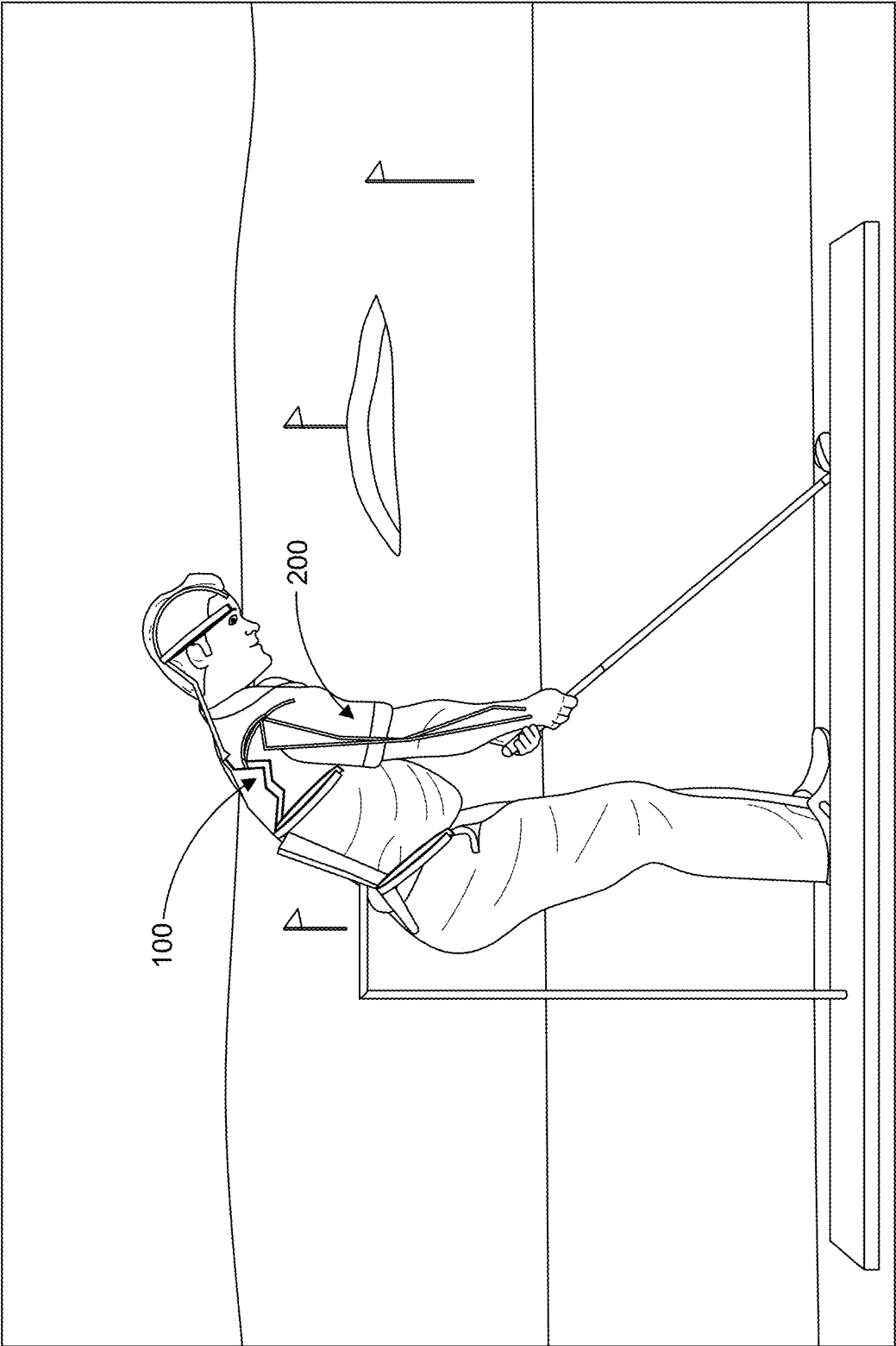


FIG. 1A

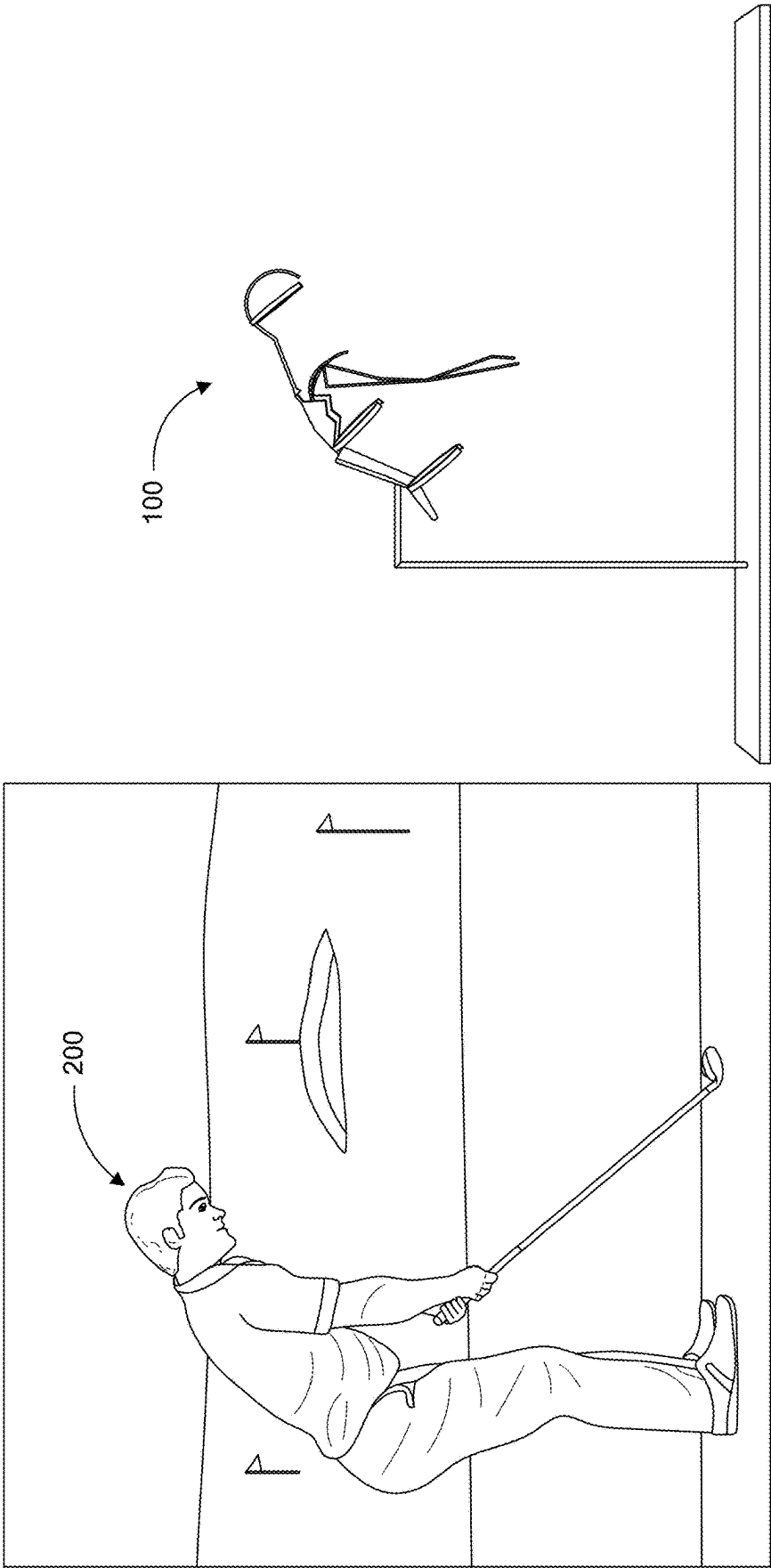


FIG. 1B

FIG. 1C

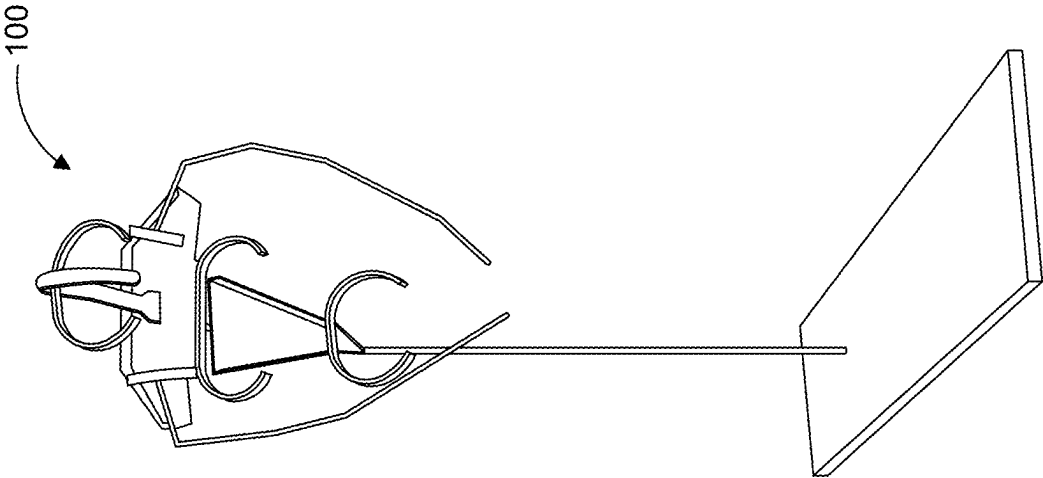


FIG. 1D

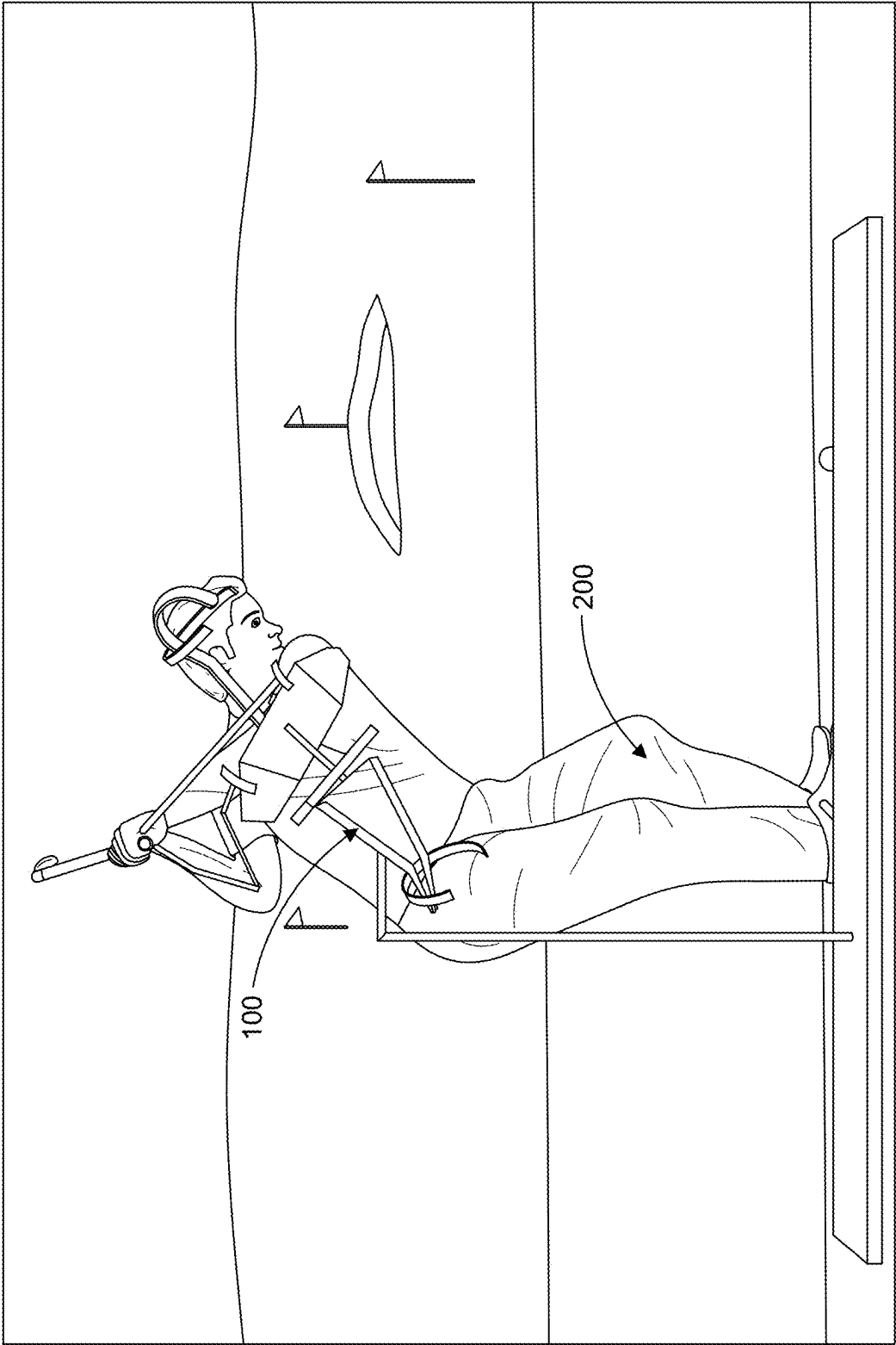


FIG. 2A

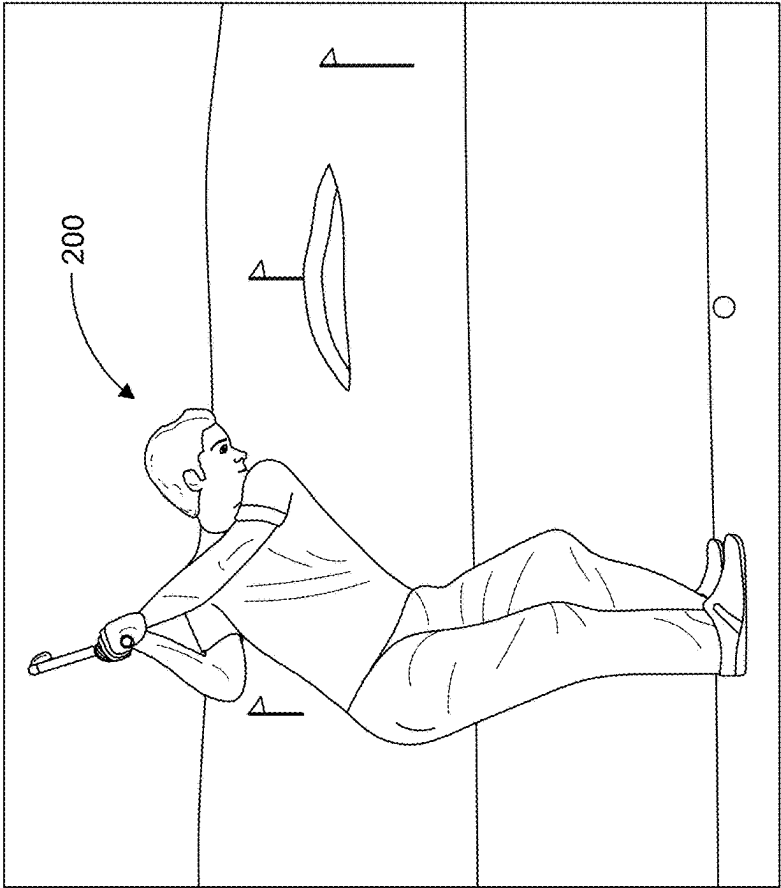


FIG. 2B

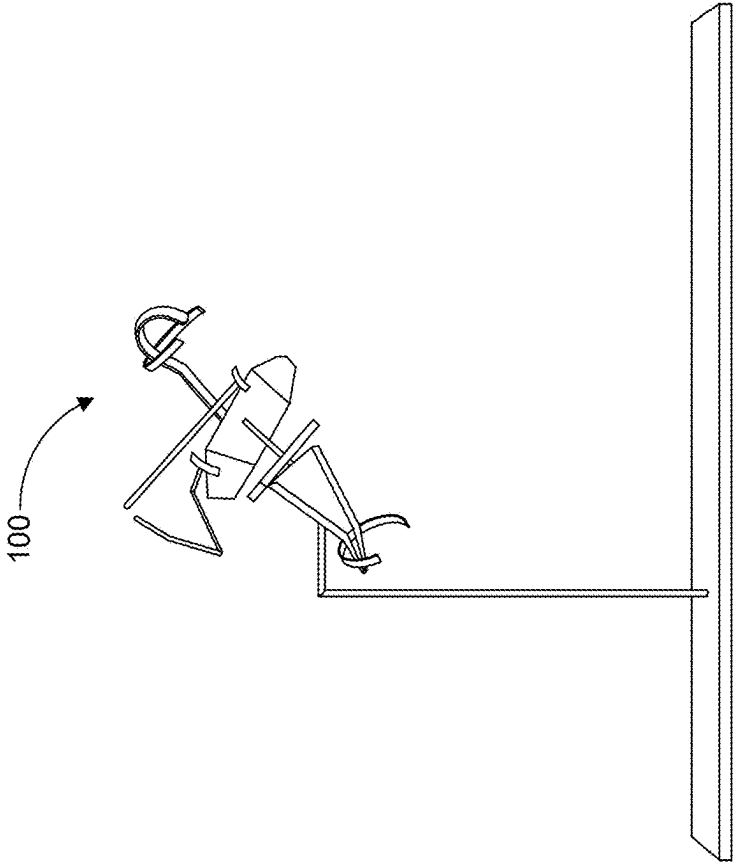


FIG. 2C

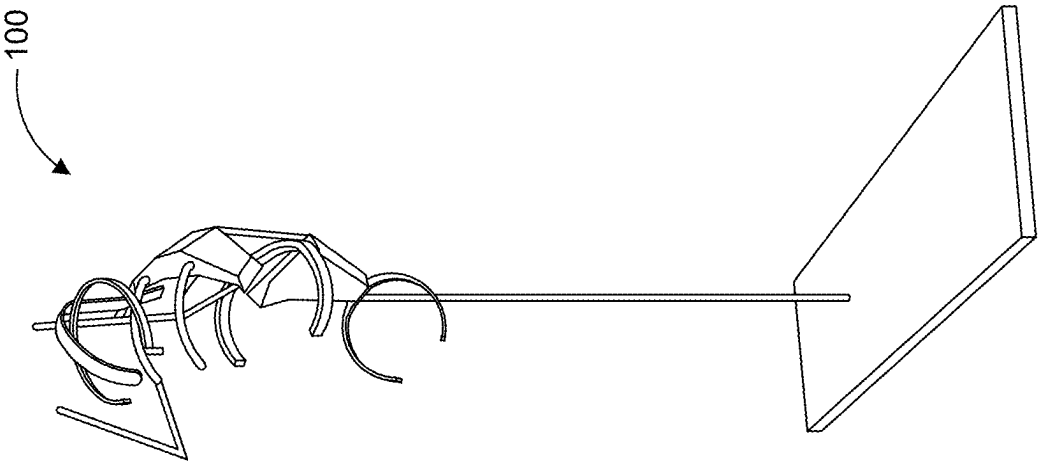


FIG. 2D

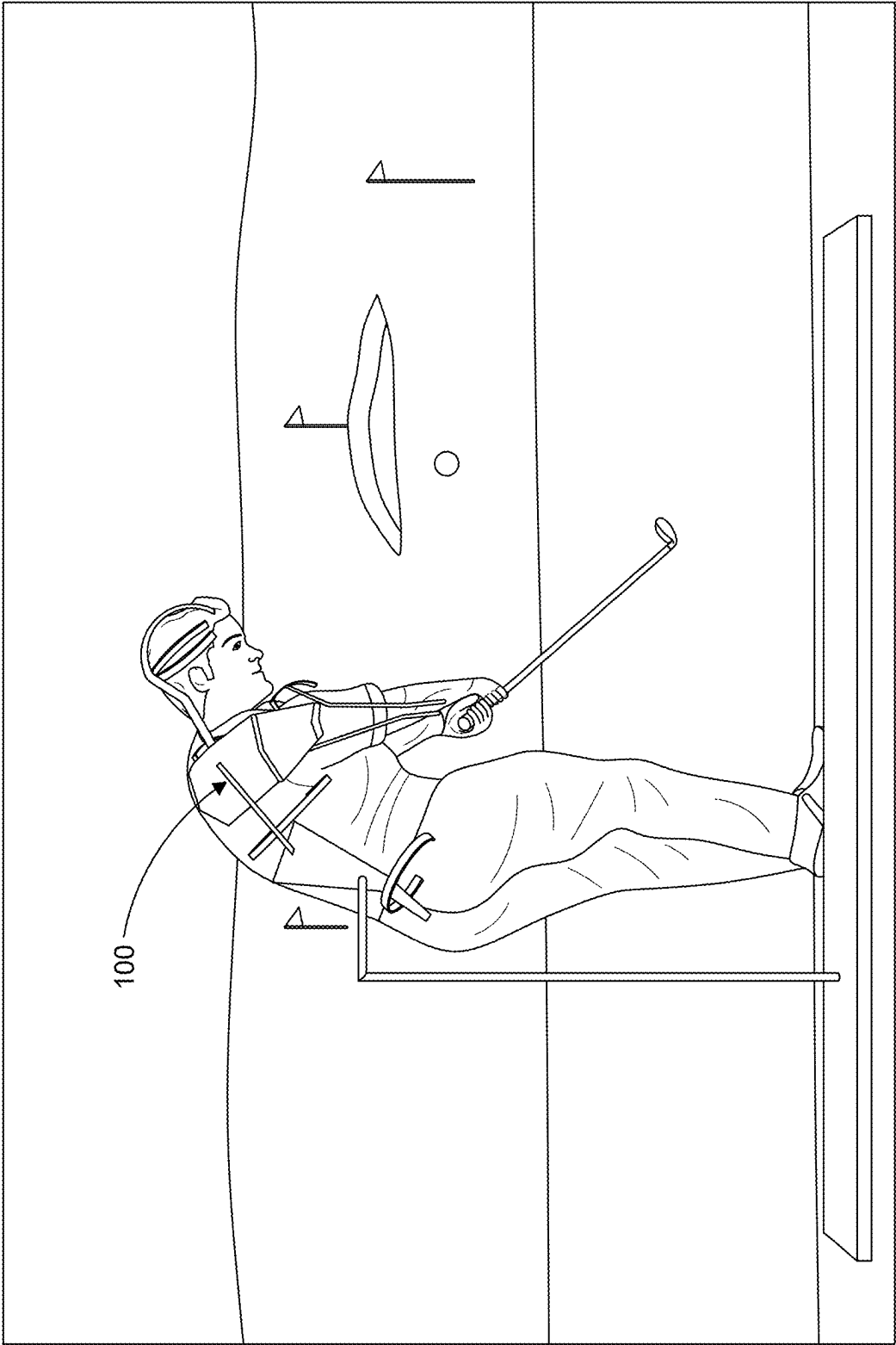


FIG. 3A



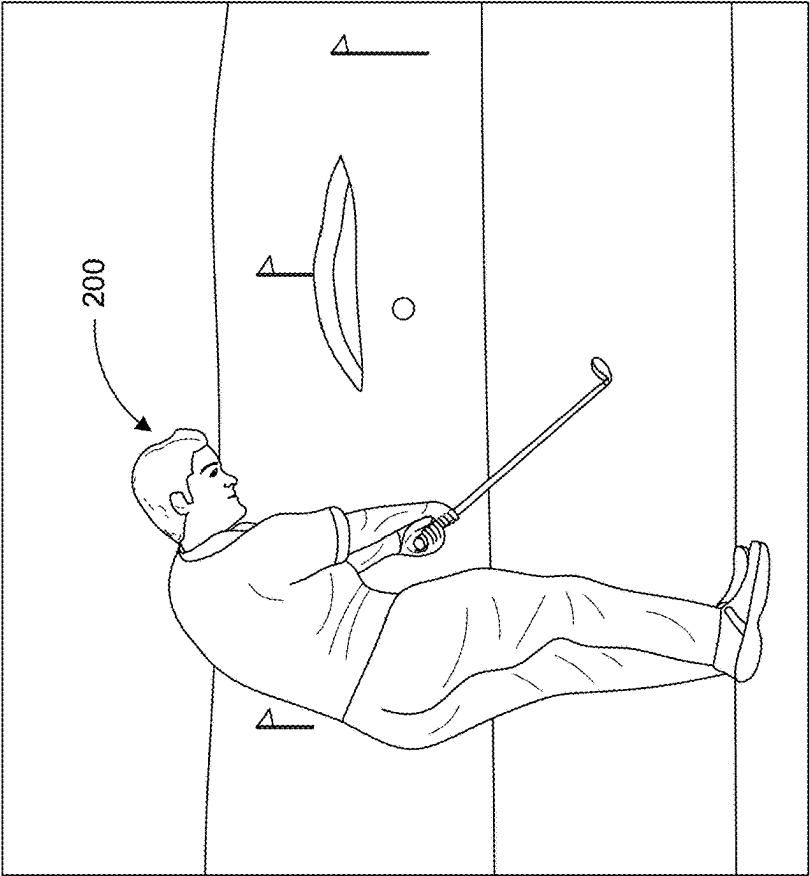


FIG. 3B

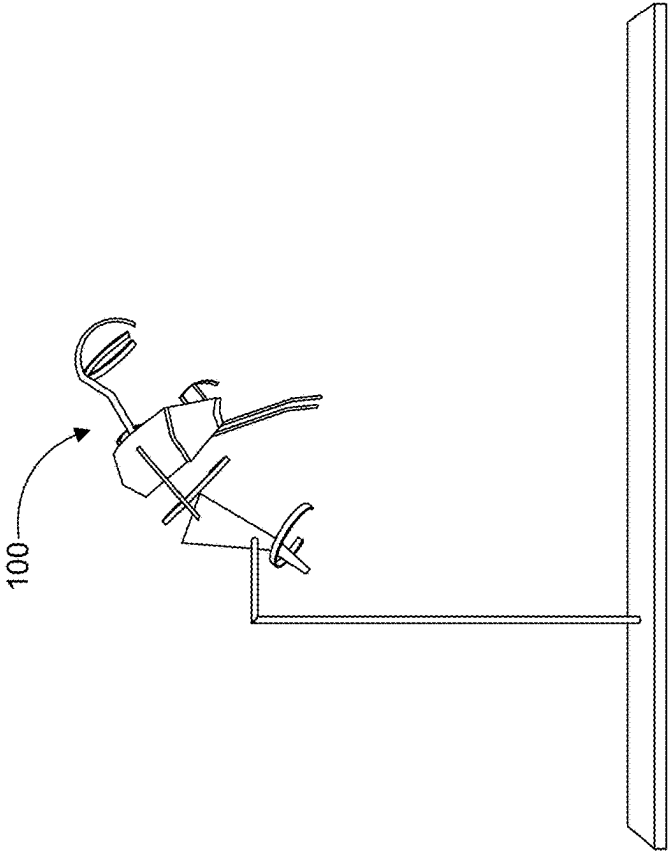


FIG. 3C

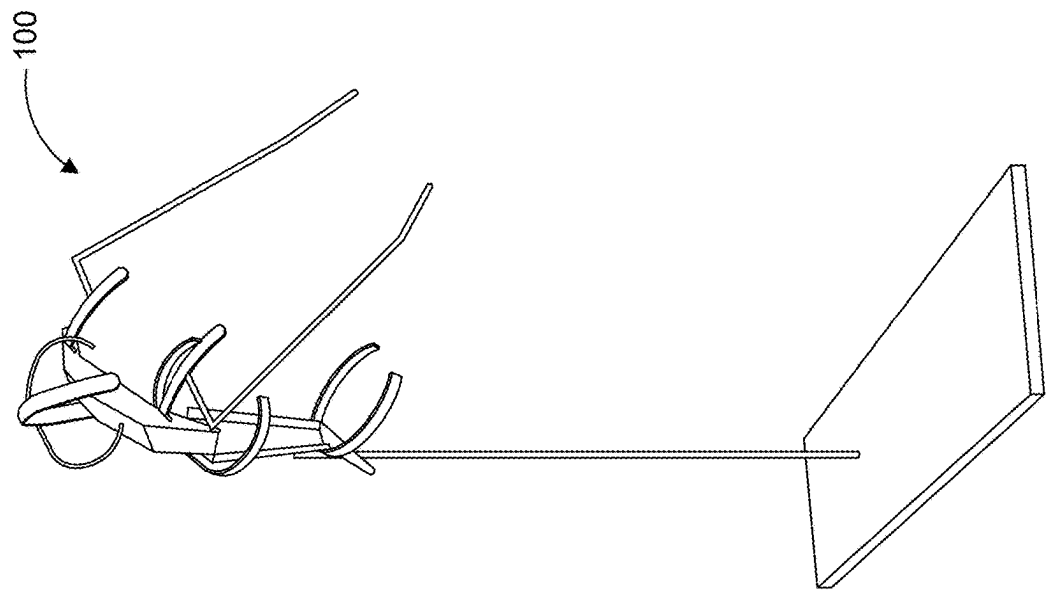


FIG. 3D

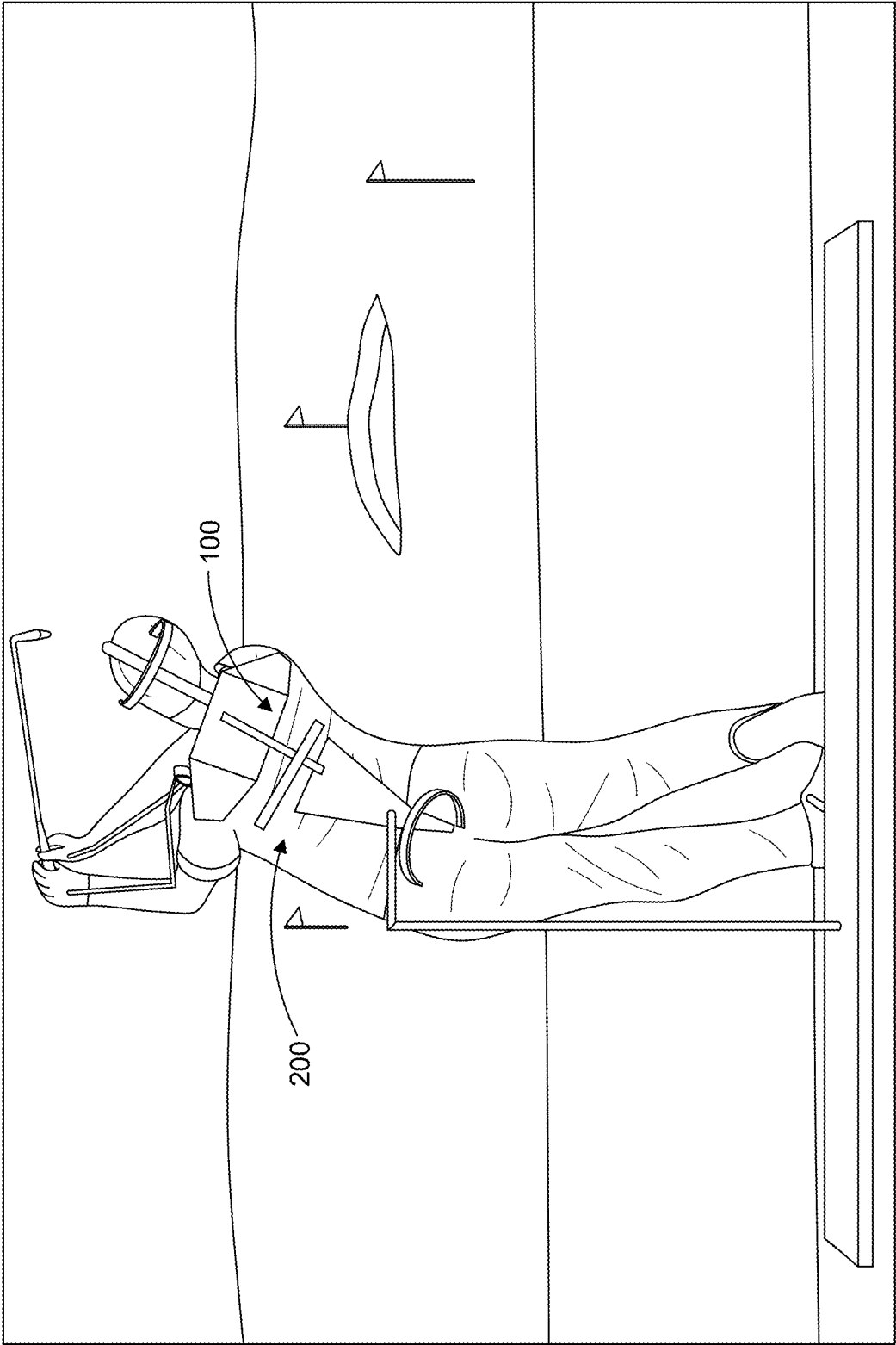


FIG. 4A

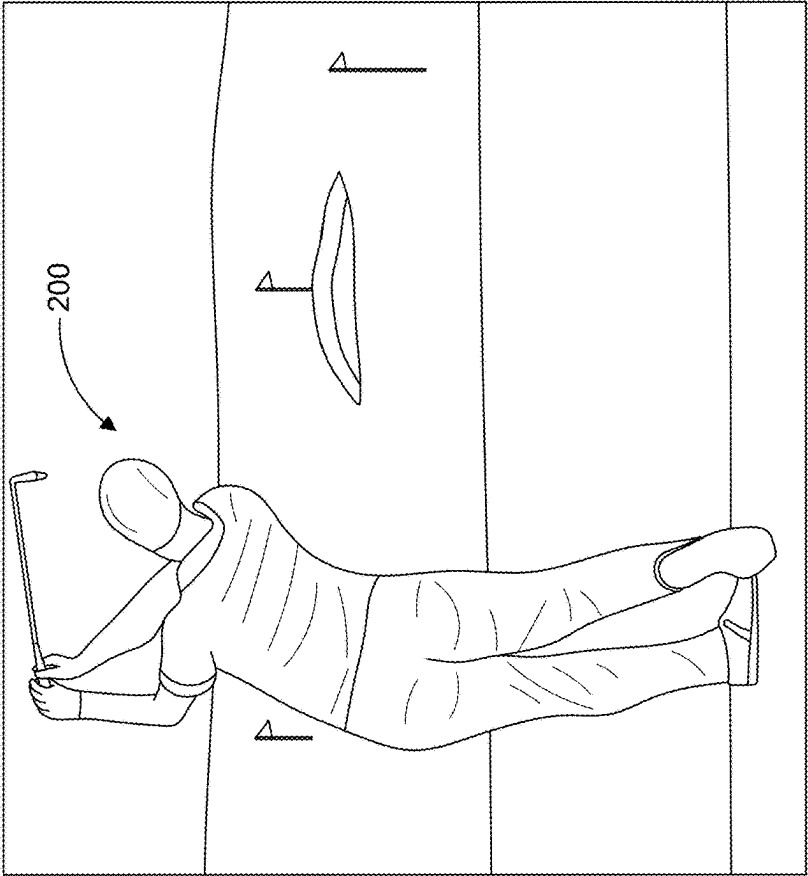


FIG. 4B

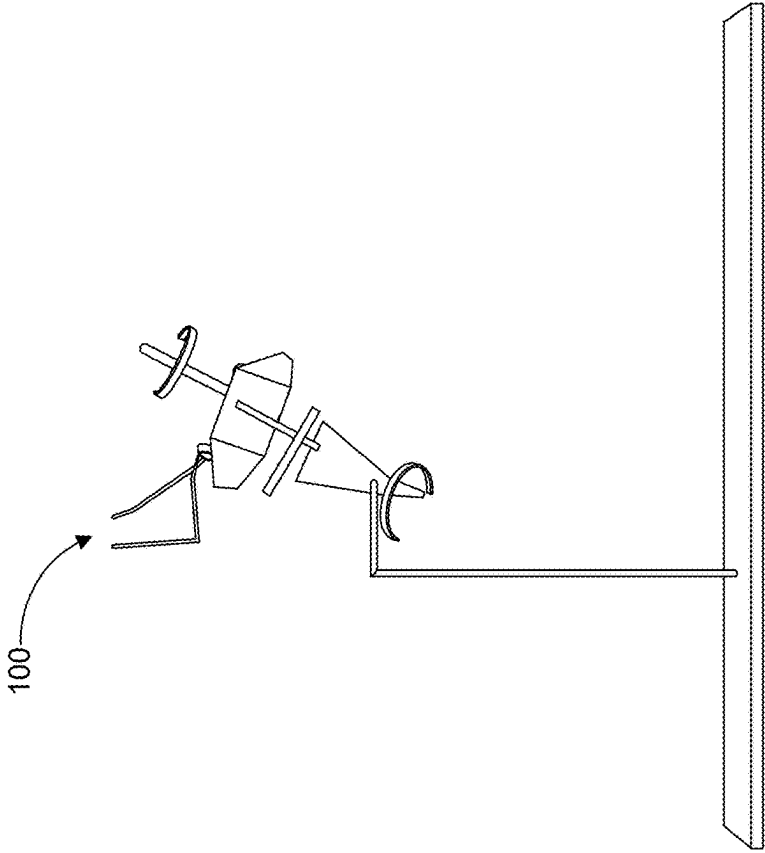


FIG. 4C

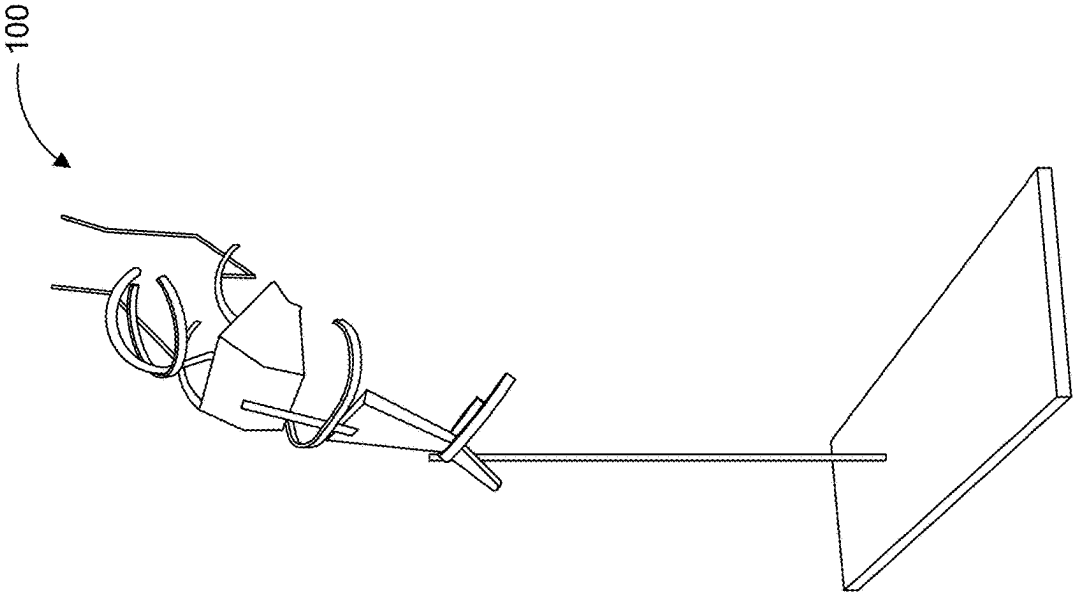


FIG. 4D

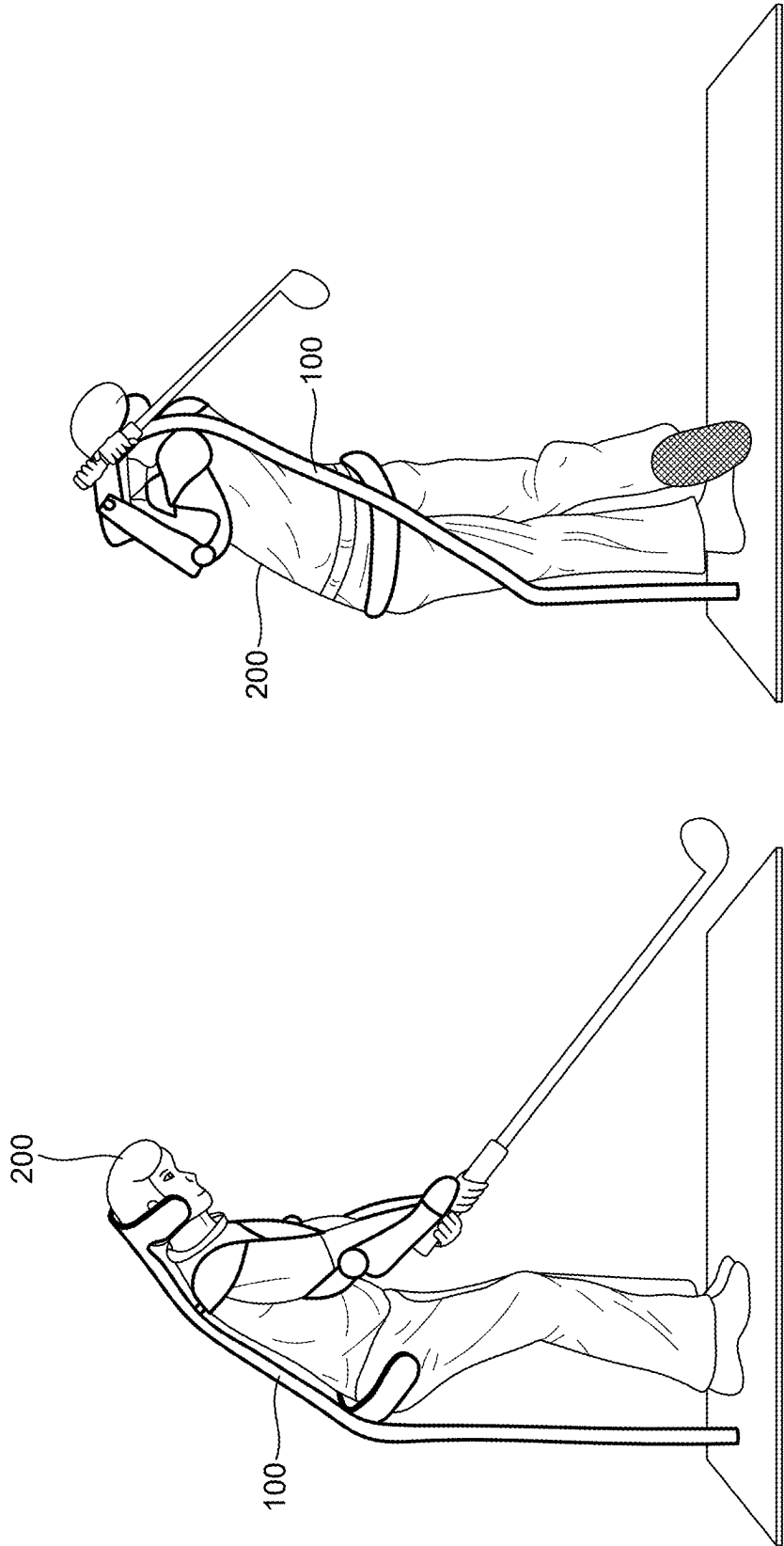


FIG. 5

## EXOSKELETON ROBOTIC SYSTEM FOR GOLF-SWING TRAINING AND MEDICAL REHABILITATION

### PRIORITY CLAIM

[0001] This application claims priority to U.S. patent application Ser. No. 18/084,377 filed Dec. 19, 2022; which application is a continuation of U.S. patent application Ser. No. 17/486,529 filed Sep. 27, 2021, which claims priority from and the benefit of U.S. Provisional Application No. 62/713,390 filed Aug. 1, 2018; the contents of which are hereby incorporated by reference in their entirety as if fully set forth herein.

### COPYRIGHT NOTICE

[0002] This disclosure is protected under United States and/or International Copyright Laws. © 2024 DAVID W. PETERSON. All Rights Reserved. A portion of the disclosure of this patent document contains material which is subject to copyright protection. The copyright owner has no objection to the facsimile reproduction by anyone of the patent document or the patent disclosure, as it appears in the Patent and/or Trademark Office patent file or records, but otherwise reserves all copyrights whatsoever.

### FIELD OF THE DISCLOSURE

[0003] The present disclosure relates generally a robotic exoskeleton and more specifically to a robotic training device and system that duplicates and guides a user to improve his/her golf swing and assists patients in rehabilitation.

### BACKGROUND

[0004] There is an insatiable demand for products and training aids to help learn a proper golf swing. All previous instruction devices, systems and methods involve an instructor guiding a student through the proper motion. Technical systems such as swing monitors and ball-flight monitors do not guide the student but simply measure results. It is difficult and for many, impossible to duplicate the instructions of a teacher and adopt their directions into one's own swing. This is how bad habits are formed which are often difficult to reverse. It is equally difficult to verbalize instructions in to a practice regimen. Accordingly there is a need for such a robotic system, device and method for golfers. Similarly, there is a need for robotic systems, devices and methods to assist patients in rehabilitation from injuries, stroke, chronic illness or other conditions involving muscular atrophy.

### SUMMARY OF THE INVENTION

[0005] A robotic device, system and method comprising at least one from the group of robotic exoskeletons for improving a golf swing, and for assisting patients in rehabilitation.

### BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0006] Preferred and alternative examples of the present invention are described in detail below with reference to the following drawings:

[0007] FIGS. 1A-1D illustrates multiple viewpoints of the robotic exoskeleton at start of swing;

[0008] FIGS. 2A-2D illustrates multiple viewpoints of the robotic exoskeleton at peak of swing;

[0009] FIGS. 3A-3D illustrates multiple viewpoints of the robotic exoskeleton at follow through point of swing;

[0010] FIGS. 4A-4D illustrates multiple viewpoints of the robotic exoskeleton at end of swing; and,

[0011] FIG. 5 illustrates an example of the exoskeleton on a subject.

### DETAILED DESCRIPTION

[0012] The invention may instruct and guide a student through a complex interconnected series of motions that produce an ideal golf swing to improve skill and enhance enjoyment of the game of golf. It may also assist patients in rehabilitation from injuries, stroke, chronic illness or other conditions involving muscular atrophy.

[0013] A robotic training device **100** that duplicates and guides a golfer **200** through a perfect golf swing in several modes. It is an exoskeleton-like robot that employs a combination of either pneumatic, electric or Bowden cables with small motors and actuators, which embraces the golfer so he/she can experience the “feel” and body movements that make a perfect golf swing and learn to perform the motion without the training device. The swing motion of a professional golfer or teaching pro can be captured by the software component of the exoskeleton robot (the “record” mode), which can then be used as the model for training students (the “play” mode). The robot can also operate independently from the user so the user can observe the motion which he will try to emulate in the training mode. The device **100** includes a tubular structure constructed of light weight metal alloy or carbon fiber, and carbon fiber arm “sleeves” and gloves. See figures. Small motors and actuators controlled by software effectuate the robot's motion.

[0014] The invention could be adopted by equipment manufacturers, golf training service providers and rehabilitation machine equipment manufacturers such as Nike, Titleist, Golftec, TrackMan Golf, Medtronic, Stryker etc.

[0015] The invention may instruct and guide a student through a complex interconnected series of motions that produce an ideal golf swing to improve skill and enhance enjoyment of the game of golf. It may also assist patients in rehabilitation from injuries, stroke, chronic illness or other conditions involving muscular atrophy.

[0016] There is an insatiable demand for products and training aids to help learn a proper golf swing. All previous instruction methods involve an instructor guiding a student through the proper motion. Technical systems such as swing monitors and ball-flight monitors do not guide the student but simply measure results. It is difficult and for many, impossible to duplicate the instructions of a teacher and adopt their directions into one's own swing. This is how bad habits are formed which are often difficult to reverse. It is equally difficult to verbalize instructions in to a practice regimen.

[0017] The robotic trainer **100** teaches several basic swing shots:

- [0018] 1. Full swing
- [0019] a. Draw
- [0020] b. Straight
- [0021] c. Cut

- [0022] 2. Knock down
- [0023] a. Draw
- [0024] b. Straight
- [0025] c. Cut
- [0026] 3. Pitch
- [0027] a. Draw
- [0028] b. Straight
- [0029] c. Cut
- [0030] 4. Chip
- [0031] a. Draw
- [0032] b. Straight
- [0033] c. Cut
- [0034] 5. Sand/Bunker
- [0035] 6. Putting
- [0036] The robot **100** can execute all shots in several speeds:
  - [0037] 1. Full speed
  - [0038] 2.  $\frac{3}{4}$  speed
  - [0039] 3.  $\frac{1}{2}$  speed
  - [0040] 4. Slow motion
- [0041] The robot **100** is controlled by voice-activation software. Commands include all combinations of shot type and speed with some natural language derivatives, such as:
  - [0042] 1. Full Swing Draw full speed
  - [0043] 2. Knock down cut  $\frac{1}{2}$  speed
  - [0044] 3. Chip shot straight slow motion
- [0045] Components of the swing:
  - [0046] 1. Setup-Address
    - [0047] a. Correct body position
    - [0048] b. Correct grip and grip pressure
    - [0049] c. Head down
    - [0050] d. Eyes on ball
  - [0051] 2. The Takeaway-Backswing (right handed. Reverse for left handed swing)
    - [0052] a. Arms fully extended
    - [0053] b. Torso, shoulders and arms rotate
    - [0054] c. Head down with no movement
    - [0055] d. Eyes on ball
    - [0056] e. Hands flat at top of swing
    - [0057] f. Left shoulder touches chin
    - [0058] g. Pressure right foot
    - [0059] h. Pressure left hand to flat position and ready to start rotating over
  - [0060] 3. Release-Swing
    - [0061] a. Head down, eyes on ball throughout swing
    - [0062] b. Arms fully extended throughout swing
    - [0063] c. Pressure right foot
    - [0064] d. Left hip returns to address position
    - [0065] e. Knees bent throughout swing
    - [0066] f. Arms start to follow through swing
    - [0067] g. Left hand starts to rotate under
    - [0068] h. Right hand starts to rotate over
    - [0069] i. Accelerate through ball at impact
    - [0070] j. Hands slightly ahead of club head
    - [0071] k. Downward striking motion at impact
    - [0072] 1. Right shoulder down to look under
    - [0073] m. Spine angle maintained
    - [0074] 4. Follow Through
      - [0075] a. Arms fully extended
      - [0076] b. Right shoulder touches chin
      - [0077] c. Right hand rotated over left for draw shot
      - [0078] i. Less rotation for straight shot
      - [0079] ii. No rotation for cut shot

- [0080] d. Torso and hips follow through
- [0081] i. Belt buckle points to target
- [0082] e. Head comes up to observe shot trajectory
- [0083] f. Elbows bend
- [0084] g. When elbows fully bent, wrists bend
- [0085] h. Left leg becomes fully straight
- [0086] 5. Statue
- [0087] a. Hold the pose while observing shot trajectory
- [0088] The student **200** enters the machine (**100**), grips a club and addresses the golf ball, speaks the desired shot and says “go” to start the motion. The robot may be augmented with Virtual Reality goggles to “see” the actual shot on a selected course and hole.
- [0089] Similarly, the robot **100** can have applications to assist patients in rehabilitation from injuries, stroke, chronic illness or other conditions involving muscular atrophy. In the “record” mode, the motion that is to be re-learned or trained is captured by the physical therapist or other medical technician. The patient can then use the device in the “play” mode to employ muscle recruitment and coordination, experience the “target” motion and re-train muscles to approximate the pre-injury or normal condition. By repetition, muscle mass is regenerated and muscle memory is learned or restored.
- [0090] The robot **100** operates in three modes which the student can observe or join:
  - [0091] 1. Automatic Guide Mode
    - [0092] a. The golf student **200** yields all control by relaxing the muscles and surrenders to the motion of the robot, allowing the machine to guide the student through the swing. The student experiences the correct position of each body part: legs, hips, torso, shoulders, arms, head and hands in the correct sequence of movement.
  - [0093] 2. Semi-Automatic Guide Mode
    - [0094] a. The student **200** engages the muscles and starts a swing which drives the robot, and which causes the robot to offer some resistance and guidance where the swing is out of compliance and directs the student into compliance with the target or perfect swing.
  - [0095] 3. Free Swing/Unconstrained Mode
    - [0096] a. The student **200** is free to make a golf swing uninhibited by the robot. The robotic device measures the components of the student’s swing that are out of compliance with the target swing and measures the degree to which the student is out of compliance with visual cues and data. The student can stop at any time to achieve the correct position for any and all components of the swing and observe the light cues as he moves into correct position. (e.g. top of backswing correct position of head, arms, shoulders, hands, hips, torso)
- [0097] i. Visual Cues:
  - [0098] 1. Green Light: within 5 degrees of target
  - [0099] 2. Yellow Light: within 20 degrees of target
  - [0100] 3. Red Light: >20 degrees out of compliance
- [0101] ii. Data Capture:
  - [0102] 1. Measures and records the degree to which the student has complied with the target or the degree to which the student is out of compliance with the target swing.
  - [0103] 2. The student can analyze the data to see where improvement is needed and the degree to which progress has been made toward the perfect swing.



**[0104]** This application is intended to describe one or more embodiments of the present invention. It is to be understood that the use of absolute terms, such as “must,” “will,” and the like, as well as specific quantities, is to be construed as being applicable to one or more of such embodiments, but not necessarily to all such embodiments. As such, embodiments of the invention may omit, or include a modification of, one or more features or functionalities described in the context of such absolute terms. In addition, the headings in this application are for reference purposes only and shall not in any way affect the meaning or interpretation of the present invention.

**[0105]** Although the foregoing text sets forth a detailed description of numerous different embodiments, it should be understood that the scope of protection is defined by the words of the claims to follow. The detailed description is to be construed as exemplary only and does not describe every possible embodiment because describing every possible

embodiment would be impractical, if not impossible. Numerous alternative embodiments could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims.

**[0106]** Thus, many modifications and variations may be made in the techniques and structures described and illustrated herein without departing from the spirit and scope of the present claims. Accordingly, it should be understood that the methods and apparatus described herein are illustrative only and are not limiting upon the scope of the claims.

What is claimed is:

1. A robotic device comprising at least one from the group of robotic exoskeletons:  
for improving a golf swing, and  
for assisting patients in rehabilitation.

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